

Step 1: calibration samples

$$\mathcal{C} = \{a_i = (x_i, y_i, z_i)^\top\}_{i=1}^n$$

Step 2: gravity estimate

$$g_0 = \frac{1}{n} \sum_{i=1}^n a_i \quad s = \|g_0\|_2 \quad \hat{d} = \frac{g_0}{s}$$

Step 3: horizontal basis

$$u_x = \frac{e_x - (e_x^\top \hat{d}) \hat{d}}{\|e_x - (e_x^\top \hat{d}) \hat{d}\|_2} \quad u_y = \frac{\hat{d} \times u_x}{\|\hat{d} \times u_x\|_2}$$

Step 4: projected components

$$v(t) = \frac{a(t)^\top \hat{d}}{s} \quad h_x(t) = \frac{a(t)^\top u_x}{s} \quad h_y(t) = \frac{a(t)^\top u_y}{s}$$

Step 5: raw tilt

$$\theta_x(t) = \text{atan2}(h_y(t), v(t)) \frac{180}{\pi} \quad \theta_y(t) = \text{atan2}(h_x(t), v(t)) \frac{180}{\pi}$$

Step 6: zeroed tilt

$$\bar{\theta}_x = \frac{1}{n} \sum_{i=1}^n \theta_x(t_i) \quad \bar{\theta}_y = \frac{1}{n} \sum_{i=1}^n \theta_y(t_i)$$

$$\Theta_x(t) = \theta_x(t) - \bar{\theta}_x \quad \Theta_y(t) = \theta_y(t) - \bar{\theta}_y$$

Variables

$a_i, a(t)$	$= (x, y, z)^\top$	$\mathcal{C}$	$= \{a_i\}_{i=1}^n$
n	$=$ calibration sample count	g_0	$=$ calibration mean vector
s	$=$ counts per g	$\hat{d}$	$=$ unit down vector
e_x	$= (1, 0, 0)^\top$	e_y	$= (0, 1, 0)^\top$
u_x	$=$ horizontal x unit vector	u_y	$=$ horizontal y unit vector
$v(t)$	$=$ vertical component	$h_x(t), h_y(t)$	$=$ horizontal components
θ_x, θ_y	$=$ raw tilt angles	$\bar{\theta}_x, \bar{\theta}_y$	$=$ calibration means
Θ_x, Θ_y	$=$ zeroed tilt angles	t_i	$=$ calibration sample time